Coasters and pendulums

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use height differences to predict speed in ideal motion.

SUCCESS CRITERIA - I can...

1. Identify the relevant vertical height change.
2. Conserve energy under stated assumptions.
3. Separate path length from gravitational energy change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “pendulum” and “potential energy” to compare your model or result. Sentence starters are optional.

pendulum: an object that swings about a fixed support

potential energy: energy associated with the configuration of interacting objects

kinetic energy: energy associated with motion

01 Bring back an earlier idea

With GMm=105 J·m, move an object from r=3 to r=7 m.
Find work by gravity using U=-GMm/r.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: mgΔh = ½mv²; v = √(2 gΔh).
Point mass released from rest, no losses or rotation; Δh is the positive vertical drop.

Worked reasoning: A pendulum released from rest 0.8 m above its lowest point reaches speed √(2 gh)=4 m/s there with g=10 and negligible losses.

Your turn: Sketch the path and energy bars at the release, bottom and opposite turning point.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

Two frictionless tracks start and end at the same heights but take different paths.
Compare final speeds from rest.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Solve and justify

A coaster enters height 8 m at speed 6 m/s.
Find speed at height 3 m with no losses.
Use g=10.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A classmate uses total ramp length as h in mgh.
Explain how to identify the correct measurement.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A positive triangular pulse peaks at 60 N and lasts 0.4 s. It acts on a 4 kg cart initially at -3 m/s.
Find impulse and final velocity.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A cart begins at rest at height 5 m. Can it reach height 6 m without an external energy input? Justify.

Word help: justify = support your answer with a reason or evidence

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A frictionless coaster starts from rest at height 8 m above the bottom of a vertical loop of radius 2 m.
Use g=10.
Find speed and normal force per unit mass at the top; check contact.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3